Aakash Kumar Agarwal

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Degree	University	Stream	Year
MS by Research	IIT Bombay	Computer Science & Engineering (AI/ML)	2023 - 2026
B.Tech	Amity University	Computer Science & Engineering	2015 - 2019

PUBLICATIONS

- **Aakash Kumar Agarwal***, Moyuru Yamada*. ***Hypothesis-Driven Reasoning for Large Language Models***, Accepted at **AAAI 2026** (Main Conference). (*Equal contribution)
- **Aakash Kumar Agarwal***, Saprativa Bhattacharjee*, Pushpak Bhattacharyya. ***ReDepress: A Cognitive Framework for Detecting Depression Relapse from Social Media***, Accepted at **EMNLP MAIN 2025**. (*Equal contribution)

WORK EXPERIENCE

- **AI Engineer Intern | Fujitsu Research, Bangalore** (May 25 - July 25)
  Guide: ***Yamada Moyuru, Fujitsu Research***
  - Developed a **Hypothesis-Driven Reasoning (HDR)** framework to improve LLM reasoning via iterative hypothesis generation and verification over multimodal data.
  - Built a two-stage memory consolidator with **LangChain** and **LangGraph**, combining automatic factor extraction with iterative refinement via a generate-and-verify loop using MLLMs.
  - Introduced an *Experience-Transfer* task to test compositional generalization across unseen multi-modal contexts. Evaluated HDR on multiple multimodal LLMs (`GPT-4o`, `Qwen2.5`, `Gemma3`), achieving near-oracle performance, **improving accuracy by up to 56.7% over baselines**.
- **Application Development Associate | Accenture, Gurugram** (May 19 - July 21)
  - Developed and maintained software applications to meet client requirements and business needs.
  - Collaborated with cross-functional teams to implement new features and enhance existing functionalities.
  - Participated in the full software development lifecycle, from requirements gathering to deployment.

RESEARCH EXPERIENCE

- **AI Engineer | CFILT Lab | IIT Bombay, Mumbai** (Jan 24 - Present)
  Guide: ***Prof. Pushpak Bhattacharya, IIT Bombay***
  - Researching applications of NLP in mental health, with a focus on depression relapse prediction from social media data.
  - Investigating LLM safety under cognitive load and multilingual adversarial conditions.

AI/ML PROJECTS

- **Detecting Depression Relapse Using Cognitive Biases**
  MS Thesis | Guide: ***Prof. Pushpak Bhattacharyya*** *(June 24 - Dec 25)*
  - Predicted depression relapse from post-remission social media behavior using cognitive bias features and temporal modeling.
  - Constructed dataset from 10 years of Reddit dumps using regex filtering and multi-stage LLM-based filtering (LLaMA 3.2 3B, LLaMA 3.1 70B), followed by clinical expert annotation.
  - Engineered cognitive bias features and performed statistical analysis confirming relapse-bias correlation.
  - Trained classical ML models (KNN, RF, SVM, NN) on bias-only features, achieving best results with Random Forest; ablation studies identified memory bias as most discriminative.
  - Implemented temporal modeling with transformer model on chronologically ordered post embeddings, improving **accuracy from 0.81 to 0.88** and **F1 from 0.76 to 0.84** by integrating cognitive biases with post embeddings.
- **Cognitive Load and Multilingual Vulnerabilities in LLM Jailbreaking**
  R&D Project | Guide: ***Prof. Pushpak Bhattacharyya*** *(Jan 25 - Dec 25)*
  - Analyzed how cognitive load and adversarial context affect LLM safety in English and four Indian languages.
  - Demonstrated that adding **cognitive load reduces LLMs' attention to safety signals**, and developed four strategies that effectively increase cognitive load and bypass safety mechanisms.
- **Multimodal Emotion Recognition**
  R&D Project | Guide: ***Prof. Pushpak Bhattacharyya*** *(Jan 24 - May 24)*
  - Adapted a LLaMA-2 based 'Listen, Think, Understand (LTU)' architecture for multimodal emotion recognition using the IEMOCAP dataset.
  - Augmented text-based inputs with speech features (volume, pitch, speaking rate) to generate enriched natural language representations using Whisper Model.
  - **Fine-tuned** the model using LoRA on LLaMa-2 model, achieving **72% classification accuracy** across emotional categories, close to state-of-the-art performance.

COURSE PROJECTS

- **Personalization of LLM Based on Profession**
  Human Centered AI: From Learning Models to Responsible Systems, Spring 2025
  - Developed a profession-aware LLM personalization framework, conditioning responses on both profession and context using **132 profession-subprofession pairs** and a dataset of **150K profession-question preference pairs** for contextual alignment and ranking.
  - Fine-tuned Gemma-2-2B with LoRA, used DPO/GRPO alignment to optimize profession-conditioned responses, improving personalization quality.
- **GradSafe: Gradient-Based Jailbreak Detection in LLMs**
  Speech and Natural Language Processing and the Web, Autumn 2024
  - Developed **GradSafe**, a gradient-based jailbreak detection method achieving **>70% F1 score** on adversarial datasets by analyzing safety-critical parameters, and enhanced performance with Logistic Regression and SVM classifiers using **PCA** (**87K** $\rightarrow$ **1K** features), yielding **5-8% accuracy improvement**.
  - Curated and evaluated a dataset of 100 persuasive adversarial jailbreak prompts, testing LLaMA-2 variants and exposing alignment limitations using GradSafe.
- **Image Retrieval Using Neural Network based Hashing**
  Information Retrieval & Mining for Hypertext & the Web, Autumn 2024
  - Explored CIFAR-10 and CIFAR-100 datasets for the image retrieval task using **Locality-Sensitive Hashing (LSH)** based on Neural Networks, with image embeddings extracted from a Vision Transformer (ViT).
  - Achieved strong results with one-stage training of the hash code: **CIFAR-10 (mAP: 90%, Precision@10: 89.7%)** and **CIFAR-100 (mAP: 76%, Precision@10: 84%)** compared to Random LSH baselines.
  - Further improved performance with two-stage training of embedding and then hash code: **CIFAR-10 (mAP: 96%, Precision@10: 95.5%)** and **CIFAR-100 (mAP: 87%, Precision@10: 84%)**.
- **Explainable Malware Classification**
  Foundations of Machine Learning, Autumn 2023
  - Built **ResNet-50** and **EfficientNetV2** for multiclass malware detection by converting binaries to grayscale images, using Microsoft Malware Classification Challenge and Mali-MG datasets, achieving **97.25% / 94.85%** (Microsoft) and **98.71% / 97.58%** (Mali-MG), with ResNet-50 matching **state-of-the-art performance**.
  - Applied **SHAP values** to the models for interpretability of **feature-level insights** into malware classification.

TECHNICAL SKILLS

- **AI & ML:** Transformers, LLMs, Fine-tuning, LoRA, Prompt Engineering, RAG (Vector Database), Safety Alignment, Hugging Face, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Langchain, AI Agents (MCP, Tool Use)
- **Programming Languages:** Python, C++
- **Web & Database:** SQL, HTML, CSS
- **Tools & DevOps:** Shell, Git, GitHub, Docker, FastAPI

KEY COURSES TAKEN

- Speech and Natural Language Processing and the Web
- Deep Learning for Natural Language Processing
- Information Retrieval & Mining for Hypertext & the Web
- Human-Centered AI
- Foundations of Machine Learning
- Automatic Speech Recognition
- Algorithms and Complexity
- Software Lab

POSITIONS OF RESPONSIBILITY

- **Co-Lead | Inter-IIT Tech Meet 14.0** *(Oct 25 - Dec 25)*
  - Co-led a 20-member team to develop a real-time Dynamic Trust Profile system integrating KYC, credit risk, and fraud analytics.
  - Designed core architecture across streaming pipelines, incremental ML, and LLM-driven decisioning for financial intelligence.
- **Teaching Assistant | Organisation of Web Processing** (Prof. Soumen Chakrabarti) *(Aug 24 - Dec 24)*
  - Assisted in invigilation, grading, and exam management to ensure smooth course execution.
- **Teaching Assistant | Computer Programming and Utilization** (Prof. Shivaram) *(Jan 24 - May 24)*
  - Collaborated with a team of 50+ TAs to support a class of 700+ students.
  - Supervised lab sessions, conducted help sessions, and fostered a productive learning environment.

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 156** in **GATE Computer Science 2023** among **75,680 candidates**

INTERESTS AND HOBBIES

- Programming, Badminton, Table Tennis, Chess